

Chapter 16: Thermodynamics

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CHM111
Lecture 16



Chapter 16: Thermodynamics

Thermodynamics extends our study of energy (Ch. 5) to answer a new question:
Which processes actually happen on their own?

- ▶ Chapter 5 asked: *how much energy* is exchanged?
- ▶ Chapter 16 asks: *which direction* does a process go?
 - ▶ The answer requires two quantities — **entropy** (S) and **free energy** (G).

Chapter Overview

Topics Covered in Chapter 16

1. **Spontaneity** — direction of change; dispersal of matter and energy
2. **Entropy** (S) — microstates; Boltzmann's equation $S = k \ln W$
3. **The Second and Third Laws** — $\Delta S_{\text{univ}} \geq 0$; standard entropies
4. **Free Energy** (G) — $\Delta G = \Delta H - T\Delta S$; equilibrium and temperature dependence

Part I

Spontaneity

Outline of Part I: Spontaneity

Spontaneous and Nonspontaneous Processes

Outline of Part I: Spontaneity

Spontaneous and Nonspontaneous Processes

What Is a Spontaneous Process?

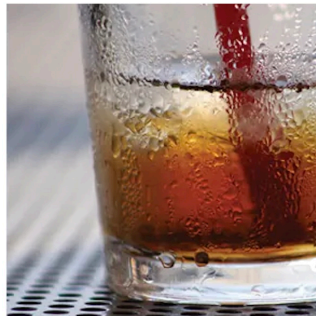
Definition: Spontaneous Process

- ▶ A **spontaneous process** occurs *without any continuous outside intervention*.
 - ▶ A **nonspontaneous process** requires a *sustained input of energy* from the surroundings to take place.
-
- ▶ A rock rolls downhill *spontaneously*; it does not roll uphill by itself.
 - ▶ Hot coffee cools in a room; it never warms itself back up.
 - ▶ Iron rusts in moist air; rust does not un-rust into iron and oxygen.

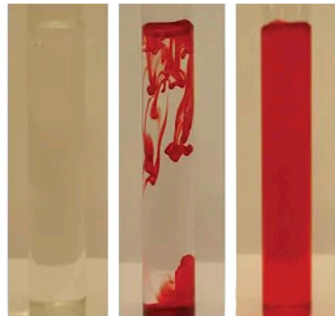
Everyday Spontaneous Processes



(a)



(b)



(c)

(a) An ice cube melts in a warm room. (b) A hot beverage cools toward room temperature. (c) A drop of dye disperses through water. Each occurs **on its own**; the reverse would demand outside intervention.

Spontaneous $\neq$ Fast

Common Misconception

“Spontaneous” does **not** mean fast. It only means that a process *tends* to proceed in a certain direction *on its own*.

Thermodynamics vs. Kinetics

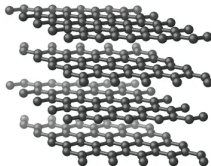
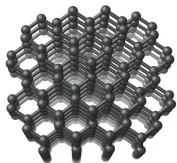
- ▶ **Thermodynamics** tells us *whether* a process is favorable.
- ▶ **Kinetics** (Ch. 12) tells us *how fast* it happens.
- ▶ A process may be spontaneous yet immeasurably slow at room temperature.

Diamond → Graphite: Spontaneous but Slow

C (diamond)



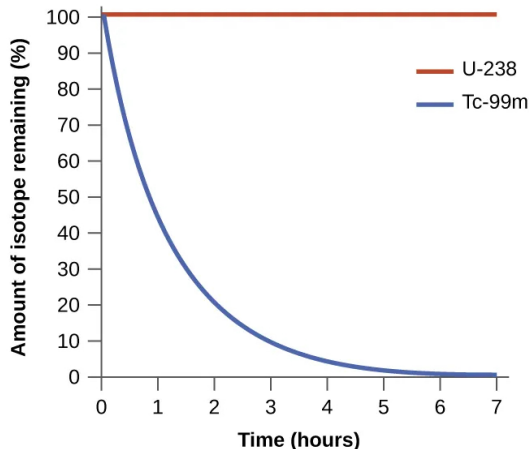
C (graphite)



The conversion of **diamond** → **graphite** at 1 atm and 25 °C is **spontaneous** — but takes billions of years.

- ▶ Graphite is the more stable allotrope of carbon at STP.
- ▶ Diamond is only *kinetically* trapped by a huge activation barrier.
- ▶ “Diamonds are forever” is kinetics, not thermodynamics.

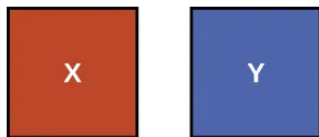
Rate v. Spontaneity: Isotope Decay



Both nuclear decays are **spontaneous**.
Only *how fast* they proceed differs.

- ▶ Tc-99m: half-life ≈ 6 h (fast).
- ▶ U-238: half-life $\approx 4.5 \times 10^9$ yr (slow).
- ▶ Same word, wildly different timescales.

Heat Flows Spontaneously from Hot to Cold



$$T_X > T_Y$$

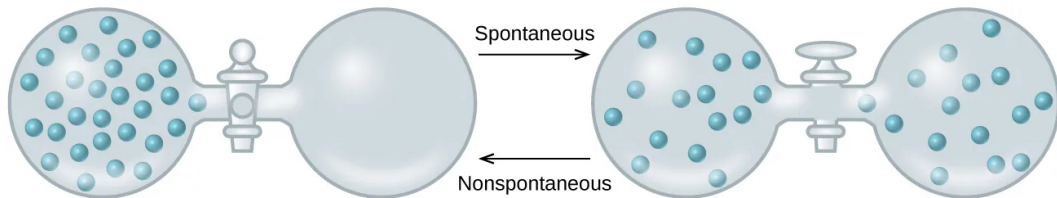


X and Y in contact

When two objects at different temperatures are placed in thermal contact, heat flows **spontaneously** from the hot object (X) to the cold object (Y) until $T_X = T_Y$.

The reverse — heat flowing from cold to hot on its own — is **never observed**.

Dispersal of Matter: Free Expansion



Opening the stopcock lets an ideal gas spontaneously spread throughout both flasks. No energy is exchanged with the surroundings ($\Delta U = 0$), yet the process proceeds — **matter has become more dispersed.**

What Drives Spontaneous Change?

Two Kinds of Dispersal

- ▶ **Matter** spreads out (gases fill their containers, solutes mix).
- ▶ **Energy** spreads out (hot bodies cool, kinetic energy is shared).

Neither dispersal violates the First Law: energy is still conserved.
But the First Law alone cannot explain *direction*.

We need a new state function — **entropy** (S) — that measures dispersal.

Check Your Understanding

A cup of hot cocoa sits on a table and cools to room temperature.

- (a) Is the cooling process spontaneous?
- (b) Would the reverse process ever occur on its own?

Come to lecture to see the answer.

Check Your Understanding

Diamond is thermodynamically unstable relative to graphite at room temperature. Why do we still see diamonds in nature and in jewelry stores?

Come to lecture to see the answer.

Part II

Entropy

Outline of Part II: Entropy

Entropy and Microstates

Predicting the Sign of Entropy Change

Outline of Part II: Entropy

Entropy and Microstates

Predicting the Sign of Entropy Change

Entropy (S) — SI Unit: J/K

Definition

Entropy is a state function measuring the number of microscopic arrangements (W) available to a system:

$$S = k_B \ln W$$

where $k_B = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$ is Boltzmann's constant.

- ▶ More accessible arrangements $\Rightarrow$ higher entropy.
- ▶ Systems tend toward states with the most microstates.
- ▶ Entropy is a *measure of dispersal*, not “disorder” in the sloppy sense.

Microstates vs. Macrostates

Microstate

A specific arrangement of every particle — positions, momenta, orientations.

A microscopic “snapshot” of the system.

A macrostate with more microstates (W) is **overwhelmingly more likely** to be observed.

Macrostate

The state characterized by macroscopic variables (P , V , T , n).

Many microstates share the same macrostate.

Counting Microstates: Particles in Two Boxes



Four particles are distributed between two connected boxes. The **even split (c)** has the greatest number of microstates (6 of 16 total), so it is the most probable macrostate. Extremes (a, e) have only 1 microstate each.

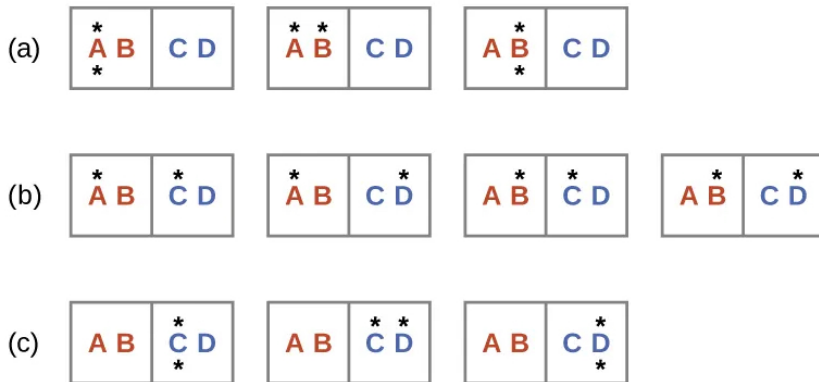
Probability of Each Distribution

Four Distinguishable Particles in Two Boxes

$$P(\text{even split}) = \frac{6}{16} = 37.5\%, \quad P(\text{all one side}) = \frac{1}{16} = 6.25\%$$

- ▶ With 4 particles, the “all on one side” outcome is rare but not impossible.
- ▶ With $N = 6.022 \times 10^{23}$ particles (1 mole), $P(\text{all on one side}) \approx 2^{-N}$ — effectively zero.
- ▶ That is why gases **never** spontaneously compress themselves into one flask.

Distributing Energy Quanta Among Particles



Two energy quanta (*) are distributed among four particles (A, B, C, D) in two boxes. Configuration (b), with one quantum on each side, has the most microstates — energy is most likely to be **dispersed** across both boxes.

Entropy Change ΔS

Change in Entropy

$$\Delta S = S_f - S_i = k_B \ln \frac{W_f}{W_i}$$

- ▶ $\Delta S > 0$: system has more microstates in the final state (more dispersed).
- ▶ $\Delta S < 0$: system has fewer microstates in the final state (more concentrated).
- ▶ $\Delta S = 0$: same number of microstates.

Entropy is a **state function**: ΔS depends only on initial and final states.

Outline of Part II: Entropy

Entropy and Microstates

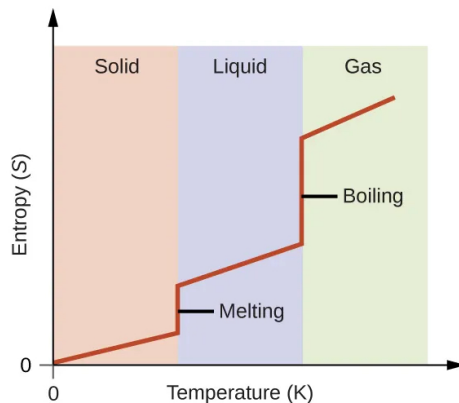
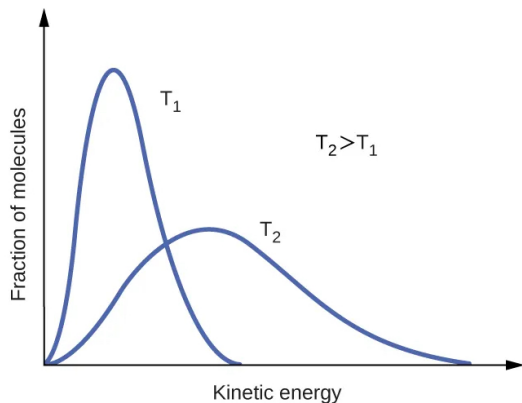
Predicting the Sign of Entropy Change

Factors That Increase Entropy

Entropy of a substance **increases** when:

- ▶ Temperature rises (more thermal microstates accessible).
- ▶ Volume increases (more positional microstates).
- ▶ Phase changes: solid $\rightarrow$ liquid $\rightarrow$ gas.
- ▶ Moles of gas particles increase in a reaction.
- ▶ A pure solid or liquid dissolves in a solvent.
- ▶ A complex molecule replaces simpler ones (more vibrational modes).

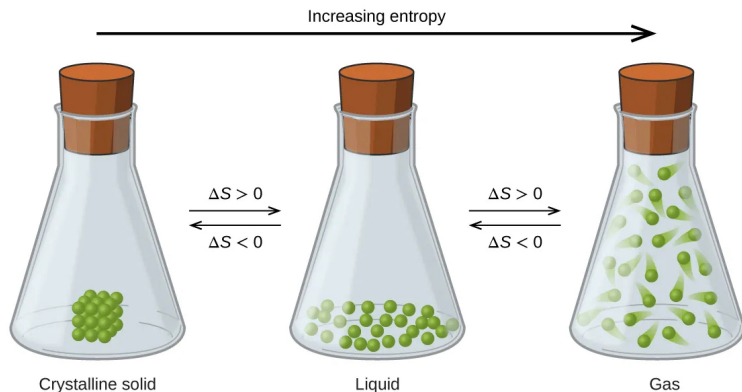
Temperature Dependence of Entropy



Left: At higher T , molecular kinetic energies are spread over a wider range of values — more thermal microstates are accessible.

Right: S rises with T , with sharp jumps at melting and boiling.

Phase Changes and Entropy



Entropy increases dramatically going solid $\rightarrow$ liquid $\rightarrow$ gas: particles gain translational and rotational freedom, occupying vastly more microstates.

Reverse processes (condensation, freezing) have $\Delta S < 0$.

Predicting ΔS for Physical Changes

- ▶ Melting, vaporization, sublimation: $\Delta S > 0$.
- ▶ Freezing, condensation, deposition: $\Delta S < 0$.
- ▶ Heating a substance: $\Delta S > 0$.
- ▶ Compressing a gas: $\Delta S < 0$.

For a chemical reaction, look first at the **change in moles of gas**:

$$\Delta n_{\text{gas}} > 0 \Rightarrow \Delta S > 0; \quad \Delta n_{\text{gas}} < 0 \Rightarrow \Delta S < 0.$$

Example: Predicting the Sign of ΔS

Sign of ΔS

Predict the sign of ΔS for each process:

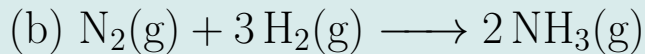
- (a) One mole liquid water boils to steam at 100 °C.
- (b) $\text{CO}_2(\text{s}) \longrightarrow \text{CO}_2(\text{g})$ (dry ice sublimates).
- (c) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2 \text{HCl}(\text{g})$.
- (d) $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2 \text{SO}_3(\text{g})$.

- (a) Liquid $\rightarrow$ gas: $\Delta S > 0$.
- (b) Solid $\rightarrow$ gas: $\Delta S > 0$.
- (c) $\Delta n_{\text{gas}} = 2 - 2 = 0$: $\Delta S \approx 0$ (small).
- (d) $\Delta n_{\text{gas}} = 2 - 3 = -1$: $\Delta S < 0$.

- Which change involves the largest positive ΔS ?
- Why is (c) close to zero even though a new substance forms?

Check Your Understanding

Predict the sign of ΔS for each process:



(c) Rubbing alcohol evaporating from your skin

Come to lecture to see the answer.

Part III

The Second and Third Laws of Thermodynamics

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

The System, the Surroundings, and the Universe

For any process we identify three regions:

- ▶ **System** — the reaction or region of interest.
- ▶ **Surroundings** — everything else that can exchange heat with the system.
- ▶ **Universe** = system + surroundings.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

The Second Law of Thermodynamics

Statement of the Second Law

For any spontaneous process, the entropy of the universe increases:

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

- ▶ $\Delta S_{\text{univ}} > 0$: process is **spontaneous**.
- ▶ $\Delta S_{\text{univ}} < 0$: process is **nonspontaneous** (the reverse is spontaneous).
- ▶ $\Delta S_{\text{univ}} = 0$: system is at **equilibrium**.

Relating Heat and Entropy

Entropy Change for a Reversible Heat Transfer

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

where q_{rev} is heat exchanged reversibly and T is absolute temperature.

 The same heat q produces a *larger* entropy change at low T .

- ▶ Adding heat to a cold system disperses energy over *few* thermal states
 - ▶ big ΔS .
- ▶ Adding heat to a hot system disperses it over *many* thermal states
 - ▶ small ΔS .

Entropy Change of the Surroundings

Key Relation

Heat that enters the system *leaves* the surroundings, so at constant P and T :

$$\Delta S_{\text{surr}} = -\frac{q_{\text{sys}}}{T} = -\frac{\Delta H_{\text{sys}}}{T}$$

- ▶ Exothermic reactions ($\Delta H_{\text{sys}} < 0$) increase ΔS_{surr} .
- ▶ Endothermic reactions ($\Delta H_{\text{sys}} > 0$) decrease ΔS_{surr} .
- ▶ The sign of ΔS_{surr} is opposite the sign of ΔH_{sys} .

Combining the Contributions: Constant T and P

ΔS_{sys}	ΔH_{sys}	Behavior
> 0	< 0	Spontaneous at all T
< 0	> 0	Nonspontaneous at all T
> 0	> 0	Spontaneous at high T
< 0	< 0	Spontaneous at low T

Temperature can **flip** the sign of ΔS_{univ} when ΔH_{sys} and ΔS_{sys} agree in sign.

Example: Is a Phase Change Spontaneous?

Melting Ice

For $\text{H}_2\text{O}(\text{s}) \longrightarrow \text{H}_2\text{O}(\text{l})$ at 1 atm: $\Delta H_{\text{fus}} = 6.00 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S_{\text{fus}} = 22.1 \frac{\text{J}}{\text{mol K}}$.

Determine whether melting is spontaneous at (a) $T = 263 \text{ K}$ and (b) $T = 283 \text{ K}$.

$$\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$(a) \Delta S_{\text{univ}} = 22.1 - \frac{6000}{263} = -\boxed{0.7 \frac{\text{J}}{\text{mol K}}}$$

$$(b) \Delta S_{\text{univ}} = 22.1 - \frac{6000}{283} = +\boxed{0.9 \frac{\text{J}}{\text{mol K}}}$$

(a) $\Delta S_{\text{univ}} < 0$: ice does **not** melt spontaneously below 273 K.

(b) $\Delta S_{\text{univ}} > 0$: ice **does** melt spontaneously above 273 K.

► Predict ΔS_{univ} at exactly 273 K.

► What does that mean physically?

Check Your Understanding

A reaction has $\Delta S_{\text{sys}} = -50 \frac{\text{J}}{\text{K}}$ and $\Delta H_{\text{sys}} = -100 \text{ kJ}$.

At $T = 298 \text{ K}$, is the reaction spontaneous?

Come to lecture to see the answer.

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

The Third Law of Thermodynamics

Statement of the Third Law

The entropy of a **perfect crystalline substance** at absolute zero is zero:

$$S(0 \text{ K}) = 0 \text{ for a perfect crystal}$$

- ▶ At 0 K, atoms have only one accessible microstate: $W = 1$, so $S = k_B \ln 1 = 0$.
- ▶ Gives entropy an **absolute** scale — unlike H and G , which only have Δ values.
- ▶ Enables tabulation of standard molar entropies S° .

Standard Molar Entropy S°

Definition

The absolute entropy of one mole of a pure substance at standard state (1 bar, specified T , usually 298 K). Units: $\text{J mol}^{-1} \text{K}^{-1}$.

- ▶ Every substance has $S^\circ > 0$ at 298 K (only 0 at 0 K).
- ▶ Gases $\gg$ liquids $>$ solids (for the same substance).
- ▶ Larger, more complex molecules have higher S° .
- ▶ Heavier atoms have higher S° than lighter atoms.

Standard Entropy Values — Some Trends

Substance	Phase	S° (J mol ⁻¹ K ⁻¹)
H ₂ O	(s)	41.0
H ₂ O	(l)	69.9
H ₂ O	(g)	188.7
CH ₄	(g)	186.4
C ₂ H ₆	(g)	229.6
Cl ₂	(g)	223.1
Br ₂	(l)	152.2
C (diamond)	(s)	2.38
C (graphite)	(s)	5.74

Notice $\text{H}_2\text{O(g)} > \text{H}_2\text{O(l)} > \text{H}_2\text{O(s)}$; graphite > diamond (looser structure); and larger molecules have larger S° .

Standard Entropy Change of Reaction, $\Delta S_{\text{rxn}}^{\circ}$

Hess-Law Style Sum

For a reaction, $\Delta S_{\text{rxn}}^{\circ}$ is the difference between the summed S° of products and reactants:

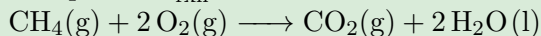
$$\Delta S_{\text{rxn}}^{\circ} = \sum n_p S^{\circ}(\text{products}) - \sum n_r S^{\circ}(\text{reactants})$$

- ▶ Weight each S° by its stoichiometric coefficient.
- ▶ Elemental S° is *not* zero (unlike ΔH_f° and ΔG_f°).
- ▶ Units: $\text{J mol}^{-1} \text{K}^{-1}$.

Example: Calculating $\Delta S_{\text{rxn}}^{\circ}$

Combustion of Methane

Compute $\Delta S_{\text{rxn}}^{\circ}$ at 298 K for:



Using S° (J/mol K): CH_4 186.4; O_2 205.2; CO_2 213.8; $\text{H}_2\text{O}(\text{l})$ 69.9.

$$\begin{aligned}\Delta S^{\circ} &= \left[S_{\text{CO}_2}^{\circ} + 2 S_{\text{H}_2\text{O}(\text{l})}^{\circ} \right] \\ &\quad - \left[S_{\text{CH}_4}^{\circ} + 2 S_{\text{O}_2}^{\circ} \right] \\ &= (213.8 + 2 \cdot 69.9) \\ &\quad - (186.4 + 2 \cdot 205.2) \\ &= 353.6 - 596.8\end{aligned}$$

$$\Delta S_{\text{rxn}}^{\circ} = -243.2 \frac{\text{J}}{\text{mol K}}$$

$\Delta S < 0$ — moles of gas decrease from 3 to 1, and one product is liquid.

- ▶ Would ΔS° be different if $\text{H}_2\text{O}(\text{g})$ were the product instead?
- ▶ Which sign is expected before doing arithmetic?

Check Your Understanding

Rank in order of *increasing* standard molar entropy:
 $\text{H}_2\text{O}(\text{s})$, $\text{H}_2\text{O}(\text{l})$, $\text{H}_2\text{O}(\text{g})$, $\text{NaCl}(\text{s})$, $\text{HCl}(\text{g})$

Come to lecture to see the answer.

Part IV

Free Energy

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

Outline of Part IV: Free Energy

Gibbs Free Energy

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Chapter Summary

Gibbs Free Energy G

Definition (J. W. Gibbs, 1876)

$$G = H - TS$$

At constant T and P ,

$$\Delta G = \Delta H - T\Delta S$$

- ▶ G combines H and S into a single spontaneity criterion.
- ▶ ΔG refers only to the **system**, unlike ΔS_{univ} .
- ▶ Units of ΔG : kJ/mol (matching ΔH).

Deriving ΔG from ΔS_{univ}

From the Second Law

Start with $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$; at constant T and P :

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T}$$

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

$$\boxed{\Delta G_{\text{sys}} = -T\Delta S_{\text{univ}}}$$

Because $T > 0$, the sign of ΔG_{sys} is *opposite* to ΔS_{univ} .

ΔG and Spontaneity

Spontaneity Criterion (constant T, P)

$\Delta G < 0$ Spontaneous (exergonic)

$\Delta G > 0$ Nonspontaneous (endergonic)

$\Delta G = 0$ At equilibrium

- ▶ A single number (ΔG) settles direction.
- ▶ We can now stay entirely within the system — no need to compute ΔS_{surr} .
- ▶ G is the “free” energy available to do non- PV work.

Standard Free Energy of Reaction $\Delta G_{\text{rxn}}^{\circ}$

Two Equivalent Routes

$$\Delta G_{\text{rxn}}^{\circ} = \Delta H_{\text{rxn}}^{\circ} - T \Delta S_{\text{rxn}}^{\circ}$$

$$\Delta G_{\text{rxn}}^{\circ} = \sum n_p \Delta G_f^{\circ}(\text{products}) - \sum n_r \Delta G_f^{\circ}(\text{reactants})$$

- ▶ ΔG_f° = standard free energy of formation (from elements).
- ▶ $\Delta G_f^{\circ} = 0$ for a pure element in its standard state.
- ▶ The two routes give the same answer (Hess's Law generalized).

Example: Two Ways to Compute $\Delta G_{\text{rxn}}^{\circ}$

Combustion of Methane

For $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ at 298 K:

$$\Delta H_{\text{rxn}}^{\circ} = -890.4 \frac{\text{kJ}}{\text{mol}}, \Delta S_{\text{rxn}}^{\circ} = -243.2 \frac{\text{J}}{\text{mol K}}.$$

Method 1: use ΔH° and ΔS° .

$$\begin{aligned}\Delta G^{\circ} &= \Delta H^{\circ} - T\Delta S^{\circ} \\ &= -890.4 - (298)(-0.2432) \\ &= -890.4 + 72.5\end{aligned}$$

$$\Delta G^{\circ} = -817.9 \frac{\text{kJ}}{\text{mol}}$$

$\Delta G^{\circ} < 0 \Rightarrow$ combustion of methane is **spontaneous** at 298 K.

The large exothermic ΔH° overwhelms the negative ΔS° .

- ▶ How would using tabulated ΔG_f° values compare?
- ▶ Why is the entropy term written in kJ, not J?

Example: Using ΔG_f° Values

Same Reaction, Different Route

For $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ use

ΔG_f° (kJ/mol): $\text{CH}_4 = -50.5$; $\text{CO}_2 = -394.4$; $\text{H}_2\text{O}(\text{l}) = -237.1$; $\text{O}_2 = 0$.

$$\begin{aligned}\Delta G^\circ &= [\Delta G_f^\circ(\text{CO}_2) + 2 \Delta G_f^\circ(\text{H}_2\text{O}(\text{l}))] \\ &\quad - [\Delta G_f^\circ(\text{CH}_4) + 2 \Delta G_f^\circ(\text{O}_2)] \\ &= [-394.4 + 2(-237.1)] - [-50.5 + 0] \\ &= -868.6 + 50.5\end{aligned}$$

$$\Delta G^\circ = -818.1 \frac{\text{kJ}}{\text{mol}}$$

Agrees with the $\Delta H^\circ - T\Delta S^\circ$ route within rounding.

Choose whichever data are more readily available.

- ▶ Why is $\Delta G_f^\circ(\text{O}_2) = 0$?
- ▶ Is the reaction spontaneous?

Check Your Understanding

A reaction has $\Delta H^\circ = +40.0 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +50.0 \frac{\text{J}}{\text{mol K}}$.

Compute ΔG° at 298 K. Is the reaction spontaneous at this temperature?

Come to lecture to see the answer.

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

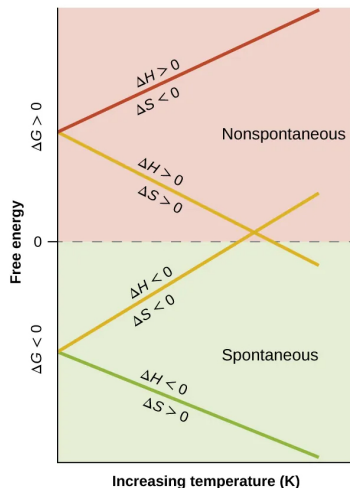
Four Combinations of ΔH and ΔS

Summary of the Four Scenarios for Enthalpy and Entropy Changes

	$\Delta H > 0$ (endothermic)	$\Delta H < 0$ (exothermic)
$\Delta S > 0$ (increase in entropy)	$\Delta G < 0$ at high temperature $\Delta G > 0$ at low temperature Process is spontaneous at high temperature	$\Delta G < 0$ at any temperature Process is spontaneous at any temperature
$\Delta S < 0$ (decrease in entropy)	$\Delta G > 0$ at any temperature Process is nonspontaneous at any temperature	$\Delta G < 0$ at low temperature $\Delta G > 0$ at high temperature Process is spontaneous at low temperature

The sign of $\Delta G = \Delta H - T\Delta S$ depends jointly on the signs of ΔH , ΔS , and on temperature.

ΔG vs. Temperature for Each Case



- ▶ **Green:** $\Delta H < 0$, $\Delta S > 0$ — always spontaneous.
- ▶ **Red:** $\Delta H > 0$, $\Delta S < 0$ — never spontaneous.
- ▶ **Yellow (up):** $\Delta H > 0$, $\Delta S > 0$ — spontaneous only above a crossover T .
- ▶ **Yellow (down):** $\Delta H < 0$, $\Delta S < 0$ — spontaneous only below a crossover T .

Crossover Temperature

At Equilibrium, $\Delta G = 0$

Setting $\Delta G = \Delta H - T\Delta S = 0$ and solving:

$$T_{\text{eq}} = \frac{\Delta H}{\Delta S}$$

- ▶ T_{eq} is the temperature at which the reaction switches direction.
- ▶ Above T_{eq} the sign of ΔG flips.
- ▶ Use ΔH and ΔS in *matching* units!

Example: Boiling Point from Thermodynamics

Boiling Point of Ethanol

For $\text{C}_2\text{H}_5\text{OH}(\text{l}) \longrightarrow \text{C}_2\text{H}_5\text{OH}(\text{g})$: $\Delta H^\circ = +42.4 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +119 \frac{\text{J}}{\text{mol K}}$.

Estimate the normal boiling point of ethanol.

$\Delta G^\circ = 0$ at boiling

$$\begin{aligned} T_{\text{bp}} &= \frac{\Delta H^\circ}{\Delta S^\circ} \\ &= \frac{42\,400 \frac{\text{J}}{\text{mol}}}{119 \frac{\text{J}}{\text{mol K}}} \end{aligned}$$

$$T_{\text{bp}} \approx 356 \text{ K} \approx 83^\circ \text{C}$$

The measured boiling point of ethanol is 78°C — a very good estimate from ΔH° and ΔS° alone.

- Why is the estimate a bit high?
- Below T_{bp} , is vaporization spontaneous?

Check Your Understanding

The reaction $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ has $\Delta H^\circ = +178 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +161 \frac{\text{J}}{\text{mol K}}$.

Above what temperature does CaCO_3 spontaneously decompose?

Come to lecture to see the answer.

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

Free Energy Away from Standard Conditions

Non-Standard Free Energy Change

For any composition, ΔG depends on the reaction quotient Q :

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where $R = 8.314 \frac{\text{J}}{\text{mol K}}$.

- ▶ ΔG° is the value at standard state ($Q = 1$).
- ▶ Q is the same reaction quotient used with K in Ch. 13.
- ▶ ΔG (not ΔG°) determines actual spontaneity.

ΔG° and K at Equilibrium

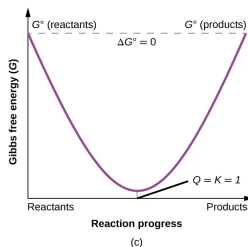
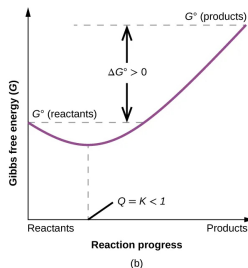
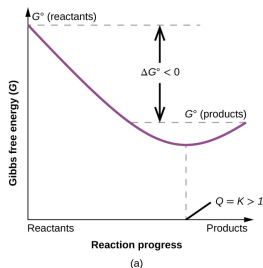
At Equilibrium, $\Delta G = 0$ and $Q = K$

$$0 = \Delta G^\circ + RT \ln K$$

$$\boxed{\Delta G^\circ = -RT \ln K}$$

- ▶ $\Delta G^\circ < 0 \Rightarrow K > 1$ (products favored).
- ▶ $\Delta G^\circ > 0 \Rightarrow K < 1$ (reactants favored).
- ▶ $\Delta G^\circ = 0 \Rightarrow K = 1$.

Reaction Progress and Free Energy



A reaction proceeds “downhill” in G to the composition where $\Delta G = 0$. The horizontal position of that minimum determines K .

(a) $\Delta G^\circ < 0 \Rightarrow K > 1$

(b) $\Delta G^\circ > 0 \Rightarrow K < 1$

(c) $\Delta G^\circ = 0 \Rightarrow K = 1$

Interpreting the Reaction Coordinate

The equilibrium composition sits at the *minimum* of G vs. progress:

- ▶ A large negative ΔG° pushes the minimum toward products.
- ▶ A large positive ΔG° pushes it toward reactants.
- ▶ Reactions do not always run “to completion” — they run to the minimum of G .

Even for very negative ΔG° , a tiny amount of reactant remains at equilibrium.

Example: ΔG° from K

Free Energy $\leftrightarrow$ Equilibrium Constant

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ at 298 K, $K_p = 0.144$.

Compute ΔG° .

$$\begin{aligned}\Delta G^\circ &= -RT \ln K \\ &= -(8.314)(298) \ln(0.144) \\ &= -(2478)(-1.938)\end{aligned}$$

$$\Delta G^\circ \approx +4.80 \frac{\text{kJ}}{\text{mol}}$$

$$K < 1 \Rightarrow \Delta G^\circ > 0.$$

The equilibrium mixture at 298 K favors N_2O_4 .

- What sign of ΔG° would $K = 1$ give?
- How could raising T shift the equilibrium here?

Check Your Understanding

For a reaction at 298 K, $\Delta G^\circ = -34.3 \frac{\text{kJ}}{\text{mol}}$.

Estimate the value of the equilibrium constant K .

Come to lecture to see the answer.

Outline of Part IV: Free Energy

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Chapter Summary

Chapter 16 Key Equations

Boltzmann's Equation

$$S = k_B \ln W$$

Reversible Heat and Entropy

$$\Delta S = q_{\text{rev}}/T$$

Surroundings

$$\Delta S_{\text{surr}} = -\Delta H_{\text{sys}}/T$$

Second Law

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

Free Energy of Reaction

$$\Delta G^\circ = \sum n_p \Delta G_f^\circ - \sum n_r \Delta G_f^\circ$$

Non-Standard ΔG

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Free Energy & K

$$\Delta G^\circ = -RT \ln K$$

Big-Picture Takeaways

- ▶ Spontaneous processes disperse **matter** and **energy**.
- ▶ Entropy S counts microstates: $S = k_B \ln W$.
- ▶ Second Law: $\Delta S_{\text{univ}} > 0$ for any spontaneous change.
- ▶ Third Law: $S = 0$ at 0 K for a perfect crystal.
- ▶ Free energy $\Delta G = \Delta H - T\Delta S$ determines whether a process is spontaneous.
- ▶ At equilibrium, $\Delta G = 0$ and $\Delta G^\circ = -RT \ln K$.

Thermodynamics tells us *whether* a reaction is thermodynamically favorable (spontaneous) by considering both energy and entropy through the Gibbs free energy, ΔG . It indicates *if a reaction can occur*.

Thermochemistry (ΔH) tells us the heat change of a reaction — whether heat is released (exothermic, $\Delta H < 0$) or absorbed (endothermic, $\Delta H > 0$), and by how much.

Kinetics tells us *how fast a reaction occurs* by examining the reaction rate and the activation energy. A reaction may be thermodynamically favorable yet still proceed very slowly if it has a large activation energy barrier.